

COMP3161 - Database Management Systems

Final Project Submission

Course Management System

Department Computing

Institution The University of the West Indies

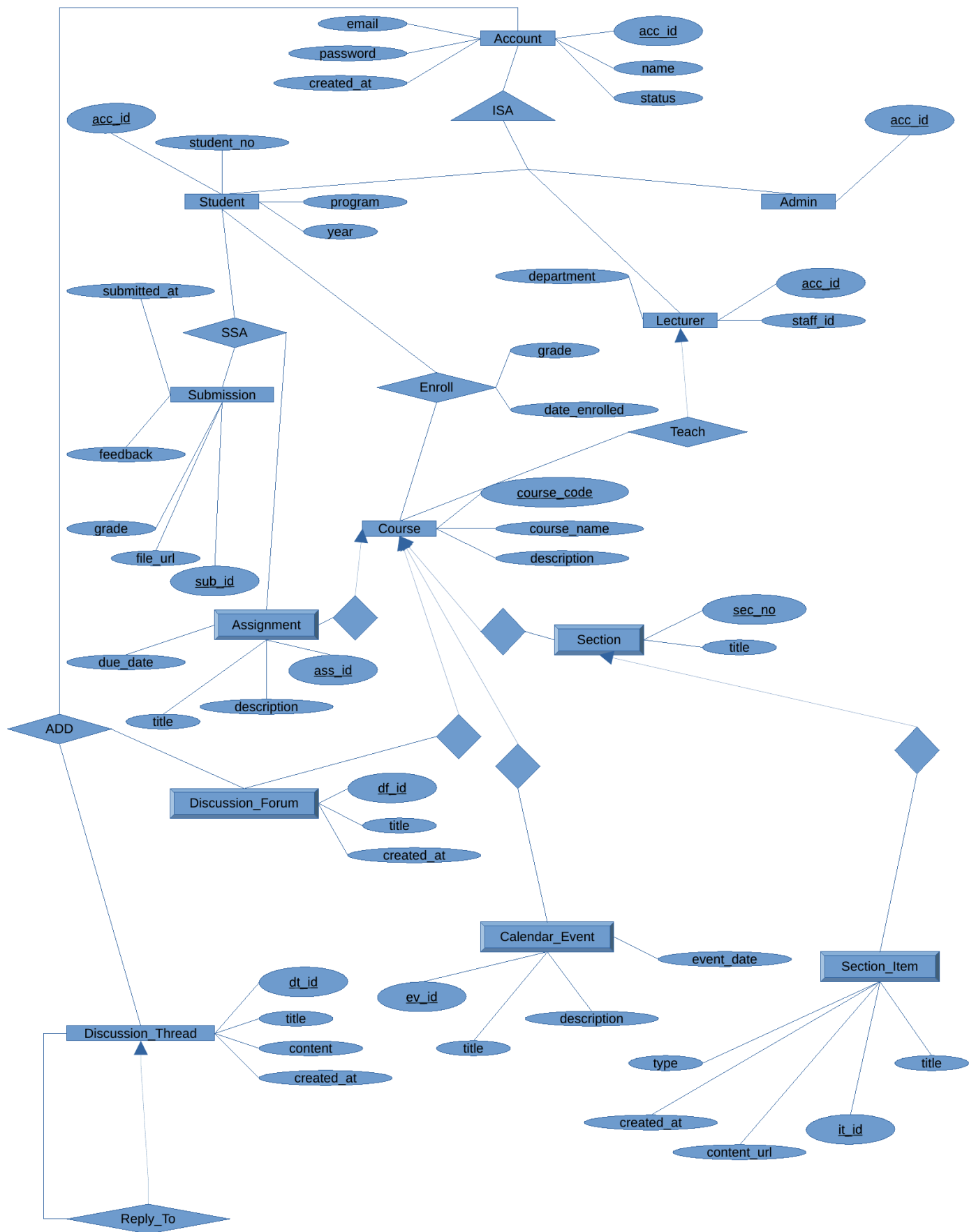
Semester Semester 2, 2025/2026

Submission Type Group Project

Group Members

Name	Student ID
Leah-Jay Holness	620163686
Alwyn Sterling	620162206
Jordayne Price	620175243
Tianna Bassaragh	620162297
Daena Crosby	620162441

Entity-Relationship Diagram



Relation Tables

Primary keys are underlined. All tables are in Third Normal Form (3NF).

Account(acc_id, name, email, password, created_at, status)

Student(acc_id, student_no, program, year)

Lecturer(acc_id, staff_id, department)

Admin(acc_id)

Course(course_code, course_name, description)

Enroll(student_id, course_code, grade, date_enrolled)

Teach(lecturer_id, course_code)

Section(sec_no, course_code, title)

Section_Item(item_id, sec_no, course_code, title, type, content_url, created_at)

Assignment(ass_id, course_code, title, description, due_date)

Submission(sub_id, submitted_at, file_url, grade, feedback)

SSA(student_id, submission_id, assignment_id)

Discussion_Forum(df_id, course_code, created_by, title, created_at)

Discussion_Thread(dt_id, title, content, created_at)

DDA(account_id, thread_id, forum_id)

Reply_To(child_id, parent_id)

Calendar_Event(ev_id, course_code, title, description, event_date)

Group Member Contributions

Leah-Jay Holness - 620163686

Role: Database Designer

- Designed the Entity-Relationship Diagram (ERD)
- Defined all entity sets, relationship sets, and attributes
- Wrote the CREATE TABLE SQL scripts with indexes
- Ensured all tables are normalized to Third Normal Form (3NF)
- Documented system assumptions and design decisions

Alwyn Sterling - 620162206

Role: Backend Developer - Authentication & Courses

- Implemented user registration and login endpoints
- Integrated JWT authentication and bcrypt password hashing
- Built course creation, retrieval, and membership management endpoints
- Implemented lecturer assignment and student enrollment logic
- Configured the psycopg2 connection pool and database layer

Jordayne Price - 620175243

Role: Backend Developer - Content, Forums & Assignments

- Built section and section item creation and retrieval endpoints
- Implemented discussion forum, thread, and reply endpoints
- Built assignment creation, submission, and grading endpoints
- Implemented calendar event creation and retrieval endpoints
- Wrote raw SQL queries for all content-related API operations

Tianna Bassaragh - 620162297

Role: Data Engineer & Reports

- Wrote the database seed script populating 100,000 students and 200 courses
- Ensured all data population constraints were satisfied
- Created the 5 required SQL report views
- Implemented Redis caching on report and course endpoints
- Performed query optimization and created database indexes

Daena Crosby - 620162441

Role: Frontend Developer & DevOps

- Built the React frontend application covering all required features
- Integrated the frontend with the REST API using Axios
- Wrote the Dockerfile and docker-compose configuration
- Set up the GitHub Actions CI/CD pipeline
- Deployed the API to Railway and the frontend to Vercel